

John Farrell

Senior Data Engineer

Marlboro, NJ | (240) 326-7636 | jwfarrell1991@gmail.com

Summary

Experienced Data Engineer specializing in designing and optimizing data pipelines, warehouses, and lakes across finance and retail domains. Skilled in building scalable, cloud-native solutions using **Google Cloud Platform (GCP)** and **Azure**, with expertise in **Apache Spark, Databricks, Google Dataflow**, and **Azure Data Factory**. Proficient in **Python, SQL, and PySpark**, with strong focus on distributed data processing, real-time streaming via **Pub/Sub** and **Kafka**, and analytics using **BigQuery**.

Skills

Programming & Scripting	Python, SQL, Java, Scala, and Shell Scripting
Big Data & Analytics	Apache Spark, Hadoop, Kafka, Databricks, Flink, Presto, Snowflake
Cloud Platforms	GCP (Big Query, Dataflow, Cloud Storage), Azure (Data Factory, Synapse, Databricks, Data Lake),
Data Engineering Tools	Apache Airflow, dbt, Apache Beam, Talend, Informatica.
Databases	PostgreSQL, MySQL, MongoDB, DynamoDB, Cassandra, Oracle
Streaming & Messaging	Kafka, Kinesis, Pub/Sub, RabbitMQ
Data Warehousing	Snowflake, Redshift, Big Query, Azure Synapse
Visualization Tools	Tableau, Power BI, Looker, Quick Sight
DevOps & Infrastructure	Docker, Kubernetes, Terraform, Ansible
Security & Compliance	IAM, KMS, Data Encryption
CI/CD & Version Control	Git, Jenkins, GitHub Actions

Work Experience

Senior Data Engineer | Nov 2022 - Present

Mastercard | New York, NY

- Designed and developed **end-to-end data pipelines** using **Google Cloud Platform (GCP)** services such as **BigQuery, Cloud Dataflow, and Cloud Composer** to support real-time and batch data processing.
- Built and optimized **enterprise-scale data lakes** on **Google Cloud Storage**, enabling seamless data ingestion and transformation for analytics and reporting.

- Developed **real-time data streaming solutions** using **Apache Kafka and Google Pub/Sub**, ensuring low-latency data processing for business intelligence applications.
- Implemented **data transformation and enrichment workflows** using **dbt (Data Build Tool) and Apache Beam**, enhancing data quality and consistency across multiple sources.
- Automated **ETL pipeline deployments and infrastructure provisioning** using **Terraform and Google Cloud Deployment Manager** for efficient cloud resource management.
- Developed **data governance and security frameworks** ensuring compliance, leveraging **Google Cloud IAM, encryption techniques, and data masking**.
- Conducted **performance tuning of SQL queries** in **BigQuery**, reducing query execution time by 40% and optimizing resource utilization.
- Developed **anomaly detection models** using **Python and TensorFlow** to identify **fraudulent activities in healthcare claims** and improve operational efficiency.
- Orchestrated **workflow automation and monitoring** with **Cloud Composer (Airflow)** to schedule and manage complex data pipelines efficiently.
- Led **migration of on-premises legacy data systems to GCP**, reducing infrastructure costs by 30% while improving scalability.
- Collaborated with **data scientists and analysts** to provide **clean, structured, and well-documented datasets** for machine learning and reporting.
- Provided **technical mentorship** to junior engineers, enforcing best practices in **CI/CD, version control (Git), and data engineering workflows**.
- Conducted **root cause analysis** and implemented **data validation frameworks** to ensure **data consistency and accuracy** across all pipelines.

Technologies Used: Google Cloud Platform (GCP), BigQuery, Cloud Dataflow, Cloud Composer, Pub/Sub, Apache Kafka, dbt, Apache Beam, Terraform, Google Cloud IAM, Python, TensorFlow, SQL, FHIR, Git, CI/CD

Data Engineer | Nov 2019 - Oct 2022

TCS | New York, NY

- Designed and developed complex **ETL pipelines** using **Azure Data Factory** to integrate data from various on-premises and cloud sources.
- Built and maintained **data lakes and data warehouses** in **Azure Synapse Analytics** to support client-specific analytical needs.
- Processed and transformed large datasets using **Apache Spark and PySpark** within **Databricks**.
- Automated data integration workflows and deployments using **Azure DevOps** for CI/CD pipelines.
- Developed interactive dashboards and reports in **Power BI** to provide actionable insights for business stakeholders.
- Containerized data processing applications using **Docker** and deployed them on **Kubernetes** for scalability and resilience.
- Implemented data ingestion pipelines to process real-time IoT data streams.
- Conducted performance tuning of **SQL Server** queries to optimize data retrieval times.
- Developed data cleansing and validation scripts in **Python** to ensure data quality across pipelines.
- Implemented role-based access control (RBAC) to manage data security and ensure compliance with industry standards.
- Collaborated with cross-functional teams to gather requirements and translate them into technical specifications.
- Migrated legacy data systems to **Microsoft Azure** for improved performance and scalability.
- Conducted root cause analysis and troubleshooting of data pipeline issues to ensure data accuracy and reliability.
- Integrated external APIs for data enrichment and transformation in Azure pipelines.
- Designed and implemented **data partitioning** strategies to improve query performance in **Azure Synapse Analytics**.

- Managed source code and version control using **Git**, ensuring code quality and versioning across teams.
- Provided technical support and training to clients on using Azure-based data solutions.
- Ensured data pipelines adhered to **GDPR** and **HIPAA** regulations for data privacy and security.
- Developed real-time data monitoring and alerting solutions using Azure services to ensure pipeline reliability.
- Led data migration projects from on-premises to cloud environments, ensuring minimal downtime and data integrity.

***Technologies Used:** Microsoft Azure, Azure Data Factory, Databricks, Azure Synapse Analytics, Apache Spark, PySpark, SQL Server, Power BI, Azure DevOps, Docker, Kubernetes, Git*

Data Engineer | Feb 2015 - Oct 2019

HCL Tech | New York, NY

- Designed and developed **end-to-end data pipelines** using **Google Cloud Platform (GCP)** services such as **BigQuery, Cloud Dataflow, and Cloud Composer** to support real-time and batch data processing.
- Built and optimized **enterprise-scale data lakes** on **Google Cloud Storage**, enabling seamless data ingestion and transformation for analytics and reporting.
- Developed **real-time data streaming solutions** using **Apache Kafka and Google Pub/Sub**, ensuring low-latency data processing for business intelligence applications.
- Implemented **data transformation and enrichment workflows** using **dbt (Data Build Tool)** and **Apache Beam**, enhancing data quality and consistency across multiple sources.
- Automated **ETL pipeline deployments** and **infrastructure provisioning** using **Terraform and Google Cloud Deployment Manager** for efficient cloud resource management.
- Developed **data governance and security frameworks** ensuring compliance with **HIPAA and GDPR regulations**, leveraging **Google Cloud IAM, encryption techniques, and data masking**.
- Conducted **performance tuning of SQL queries** in **BigQuery**, reducing query execution time by 40% and optimizing resource utilization.
- Integrated **external healthcare APIs** for patient data ingestion and processing, ensuring **FHIR (Fast Healthcare Interoperability Resources) compliance**.
- Developed **anomaly detection models** using **Python and TensorFlow** to identify **fraudulent activities in healthcare claims** and improve operational efficiency.
- Orchestrated **workflow automation and monitoring** with **Cloud Composer (Airflow)** to schedule and manage complex data pipelines efficiently.
- Led **migration of on-premises legacy healthcare data systems to GCP**, reducing infrastructure costs by 30% while improving scalability.
- Collaborated with **data scientists and analysts** to provide **clean, structured, and well-documented datasets** for machine learning and reporting.
- Provided **technical mentorship** to junior data engineers, enforcing best practices in **CI/CD, version control (Git), and data engineering workflows**.
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Education

New York Institute of Technology | Master of Science (MS) | Data Science | Sep 2013 - Jan 2015